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Studierichting: Informatica
Oefeningenreeks 3E nr. 10

$\text{Inhoud} = 360\,\text{m}^3$

$\text{grondvlak} = x^2$

$\text{hoogte} = \dfrac{360}{x^2}$

$\text{prijs grondvlak} = 400x^2$

$\text{prijs deksel} = 600x^2$

$\text{oppervlakte wanden} = 4x \cdot \dfrac{360}{x^2}$

$= \dfrac{1440}{x}$

$\text{prijs wanden} = 300 \cdot \dfrac{1440}{x}$

$= \dfrac{432000}{x}$

$p(x) = 400x^2 + 600x^2 + \dfrac{432000}{x}$

$p(x) = 1000x^2 + \dfrac{432000}{x}$

$p'(x) = 2000x - \dfrac{432000}{x^2}$

$p'(x) = 0$

$2000x - \dfrac{432000}{x^2} = 0$

$2000x^3 - 432000 = 0$

$x^3 = 216$

$x = 6$

x	0	6	$+\infty$
$p'(x)$	$-$	0	$+$
$p(x)$	$\searrow$		$\nearrow$

$p''(x) = 2000 + \dfrac{864000}{x^3}$

$p''(6) > 0$

$\Rightarrow$ minimum

$$\text{hoogte} = \frac{360}{6^2}$$

$$= \frac{360}{36}$$

$$= 10$$

$$p(6) = 1000 \cdot 36 + \frac{432000}{6}$$

$$= 36000 + 72000$$

$$= 108000$$

$$\boxed{\text{afmetingen } 6\,\text{m} \times 6\,\text{m} \times 10\,\text{m}}$$

$$\boxed{\text{minimale prijs} = 108000}$$